1. (1). First(A) = {e, b, c, d}

First(B) = {ε, b}

First(C) = {c, a, d}

First(D) = {ε, d}.

Follow(A) = {#}

Follow(B) = {a, c, d, #}

Follow(C) = {a, c, d, #}

Follow(D) = {a, c, d #}

(2) LL(1)分析表.

	a	b	c	d	e	#
A	A→BCc	A→BCc	A→BCc	A→BCc	A→eDB	A→BCC
B	B→ε	B→bCD	B→ε	B→ε		B→ε
C	C→aB		C→a	C→aB		C→aB
D	D→ε	D→ε	D→ε	D→ε, D→dB	D→ε	D→ε

不是LL(1)文法

2. (1). First (begin <stmt> end) = {begin}

First (d; <stmt>) = {d}

First (s<tail>) = {s}

First (; s<tail>) = {;}

Follow (<tail>) = {follow(<stmt>)} = {end}

	begin	end	d	s	;	#
<program>	<program> → begin <stmt> end					
<stmt>			<stmt> → d; <stmt>	<stmt> → s<tail>		
<tail>		<tail> → ε			<tail> → ; s <tail>	

(2) 分析栈

<program>

end <stmt> begin.

end <stmt>

end <stmt>; d

end <stmt>; s

end <stmt>;

end <stmt>

end <tail>

end

余符号输入串

begin d; send #

begin d; send #

d; send #

d; send #

; send #

send #

send #

end #

end #

#

产生式

<program> → begin <stmt> end

pop

<stmt> → d; <stmt>

pop

pop

<stmt> → s<tail>

pop

<tail> → ε

pop

pop

分析完成

3. (1) A → baB | ε

B → Abb | a

First(baB) = {b}

First(Abb) = {b, ε} First(a) = {a}

Follow(A) = {#, b}

Follow(B) = {#, b}

Follow(A) ∩ First(baB) = {b} ≠ ∅

不是LL(1)文法. 存在公因子 B → baBbb | bb | a

改为 A → baB | ε

B → bB' | a

B' → abbb | b.

(2) 不是LL(1)文法 存在左递归 C M → maH | H 存在公因子 (H → b(M) | (M) | b)

改为 M → HM'

M' → maHM' | ε

H → bH' | (M)

H' → (M) | ε

(3). 不是LL(1)文法存在公因子 (B → Db | D), 以及 S → AB → BaB | B.

改为 S → BS'

S' → aB | ε

B → DB'

B' → b | ε

D → d | ε

(4). S → Ab | Bb → aAb | ab | aa. 不是LL(1)文法. 存在左公因子

改为: S → aS'

S' → Ab | b | a. 仍有左公因子.

改为 S → aS'

S' → aS''b

S'' → ab | b | ε